Tour Report

Purpose: To test the Digital Teleprotection Coupler (DTPC) System between PHPA-II to Alipurduar substation line I and II.

On 10th March, 2026 our team moved from Thimphu to PHPA-II power house. We arrived at around 2:00 PM and conducted some online meeting between the officials from Power Grid Corporation of India Limited (PGCIL), officials from Tejas Network Limited and officials from Valiant Communication Limited (VCL). The main agenda in the meeting was mainly about the termination details and KLM configuration for both India and Bhutan side.

We learned during the meeting that India uses two Digital Tele-protection Couplers (DTPC) for one line as the primary DTPC and another as a backup. However, Bhutan only installed one DTPC per line, which is capable of handling both main-I and main-II relays. Following an hour-long conference, PGCIL and BPSO officials decided to modify the DTPC scheme and reorganize the KLM, as shown in the image below. We then returned to the Basochu guest house.

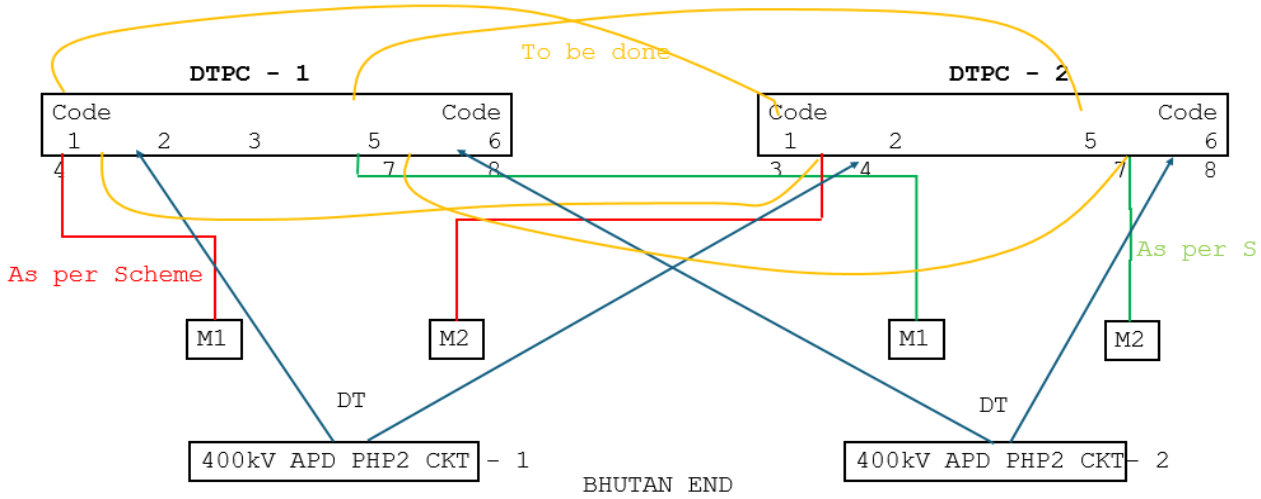

Figure 1: Above picture depicted the looping between DTPC1 and DTPC 2.

The looping between DTPC-I and DTPC-II has been done because of redundancy purpose so that each relay send/receive binary commands even if one of the DTPC failed. In addition from each relay M1/M2 can send/receive code via two channels. The yellow line above shows the looping wire recommended by the Alipurduar side.

On March 11, 2026, we completed the control cable connection and termination for the DTPC as recommended by the Indian side with the help of Mr. Rajesh from Valiant Communication Limited. By the afternoon, we received the Indian side's approval regarding the new termination prepared by the Valiant officials. At the same time, we started working on the DTPC and were able to synchronize both DTPCs on our end. However, teleprotection testing was delayed until the following day due to problems with the E1 media converter and DTPC sync. After all, we returned to the guest house in Basochu.

Following figures represents the DTPC Addressing:

Figure2: DTPC Addressing for the PHPA-II end

Figure3: DTPC Addressing for the Alipurduar end.

On March 12th, we continued to work on the sync issues caused by the E1 media converter on the India side. To address the issues, we attempted to remove the DTPC from the GIS room and relocate it to the control room, where our SDH is situated. With the help of Tejas and Valiant officials, we linked DTPC straight to the SDH without needing the E1 media converter. The results are still shown with the E1 media converter on the India side. By afternoon, the NGN's team had completed the termination and looping at the DTPC for lines I and II, according to the drawing given by Valiant officials. We returned to the Bjimethangkha DGPC guest home after the evening because Basochu's guest house was not available.

On March 13th, we proceeded to work on the sync issues by contracting Valiant officials. Even though the sync status of both DTPCs at Alipurduar was in sync, we were unable to identify the exact fault with the media converter that was flashing the LED every second. Then, as suggested by Tejas, we waited for the E1 tester to be managed by Tejas on both sides until March 15th.

\

Figure4: The above figure represented the DTPC1 and DTPC 2 for PHPA-I, Alipurduar and the Jigmeling from top to bottom.

On March 16th, we continued to investigate the issues with the E1 tester, and the results revealed that there are no issues with sending and receiving signals from DTPC to DTPC while using the E1 media converter. Finally, the Valiant and PGCIL officials suggested that we go for the Teleprotection testing despite having Media converter LED issues.

On March 17th, we discussed with PHPA-II officials the availability of a relay expert from Bhutan Automation. However, PHPA-II authorities stated that all relay expertise is currently engaged in the Yungichu's project and will not be available until late April. Then after discussion with the officials from PGCIL, Valiant, Tejas and the BPSO, decided to postpone the testing until the availability of a relay expert from Bhutan Automation.

Submitted By:

Tandin Tshewang (Engineer)

Kezang Sherab (Engineer)